Year 11 Higher

REVISION PACK

Progress Check 3

ANSWERS

Units in this pack

- Number 9
- Algebra 9

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 9 · Percentages of amounts and percentage change (Lessons 6.1, 6.2)

Q	Working	Answer
a percentage of an amount		
1.1	8.4% of $1500 = \frac{8.4}{100} \times 1500$ $= 126$	**126 litres**
1.2	3.5% of $2400 = \frac{3.5}{100} \times 2400$ $= 84$	**84 KD**
1.3	62.5% of $320 = \frac{62.5}{100} \times 320$ $= 200$	**200 students**
1.4	4.8% of $2500 = \frac{4.8}{100} \times 2500$ $= 120$	**120 litres**
1.5	12.5% of $640 = \frac{12.5}{100} \times 640$ $= 80$	**80 KD**
1.6	2.4% of $5000 = \frac{2.4}{100} \times 5000$ $= 120$	**120 grams**
a percentage of the rest		
1.7	$\text{rest} = 1500 \times 0.8 = 1200$ $\text{rent} = 1200 \times 0.3 = 360$	**360 KD**
1.8	$\text{rest} = 2400 \times 0.85 = 2040$ $\text{transport} = 2040 \times 0.3 = 612$	**612 KD**
1.9	$\text{rest} = 1600 \times 0.75 = 1200$ $\text{food} = 1200 \times 0.35 = 420$	**420 KD**
1.10	$\text{rest} = 2000 \times 0.7 = 1400$ $\text{rent} = 1400 \times 0.45 = 630$	**630 KD**

1.11	$\text{rest} = 1250 \times 0.92 = 1150$ $\text{food} = 1150 \times 0.2 = 230$	**230 KD**
1.12	$\text{rest} = 3000 \times 0.88 = 2640$ $\text{travel} = 2640 \times 0.25 = 660$	**660 KD**
percentage profit		
1.13	$\text{profit} = 37.5 - 30 = 7.5$ $\frac{7.5}{30} \times 100 = 25\%$	**25%**
1.14	$\text{profit} = 21.6 - 16 = 5.6$ $\frac{5.6}{16} \times 100 = 35\%$	**35%**
1.15	$\text{loss} = 250 - 210 = 40$ $\frac{40}{250} \times 100 = 16\%$	**16%**
1.16	$\text{profit} = 54 - 45 = 9$ $\frac{9}{45} \times 100 = 20\%$	**20%**
1.17	$\text{profit} = 75.6 - 60 = 15.6$ $\frac{15.6}{60} \times 100 = 26\%$	**26%**
1.18	$\text{loss} = 320 - 272 = 48$ $\frac{48}{320} \times 100 = 15\%$	**15%**

Number 9 · Compound interest (Lesson 6.3)

Q	Working	Answer
compound interest		
2.1	5000×1.03^3 $= 5463.64\ldots$	**5464 KD**
2.2	2800×1.015^4 $= 2971.82\ldots$	**2972 KD**
2.3	$12\,000 \times 1.045^5$ $= 14\,954.18\ldots$	**14 954 KD**

2.4	6000×1.02^5 $= 6624.48\ldots$	**6624 KD**
2.5	1800×1.035^2 $= 1928.2\ldots$	**1928 KD**
2.6	9500×1.012^3 $= 9846.12\ldots$	**9846 KD**
after how many years?		
2.7	$50\,000 \times 0.94^1 = 47$ $\ldots$ $50\,000 \times 0.94^3 = 41\,529$ $50\,000 \times 0.94^4 = 39\,037$	**4 years**
2.8	$8000 \times 0.85^1 = 68$ $\ldots$ $8000 \times 0.85^4 = 4176$ $8000 \times 0.85^5 = 355$	**5 years**
2.9	$2000 \times 1.08^1 = 216$ $\ldots$ $2000 \times 1.08^5 = 2939$ $2000 \times 1.08^6 = 3174$	**6 years**
2.10	$30\,000 \times 0.97^1 = 291$ $\ldots$ $30\,000 \times 0.97^5 = 25\,762$ $30\,000 \times 0.97^6 = 24\,989$	**6 years**
2.11	$12\,000 \times 0.9^1 = 108$ $\ldots$ $12\,000 \times 0.9^5 = 7086$ $12\,000 \times 0.9^6 = 6377$	**6 years**
2.12	$5000 \times 1.04^1 = 52$ $\ldots$ $5000 \times 1.04^4 = 5849$ $5000 \times 1.04^5 = 6083$	**5 years**

Number 9 · Reverse percentages (Lesson 6.4)

Q	Working	Answer
price before a reduction		
3.1	$85\% = 85$ $\text{original} = 85 \div 0.85 = 100$	**100 KD**
3.2	$70\% = 42$ $\text{original} = 42 \div 0.7 = 60$	**60 KD**
3.3	$92\% = 1748$ $\text{original} = 1748 \div 0.92 = 1900$	**1900 KD**
3.4	$75\% = 120$ $\text{original} = 120 \div 0.75 = 160$	**160 KD**
3.5	$90\% = 76.5$ $\text{original} = 76.5 \div 0.9 = 85$	**85 KD**
3.6	$95\% = 2850$ $\text{original} = 2850 \div 0.95 = 3000$	**3000 KD**
how much more after a rise?		
3.7	$\text{before} = 1680 \div 1.12 = 1500$ $1680 - 1500 = 180$	**180 KD**
3.8	$\text{before} = 2016 \div 1.05 = 1920$ $2016 - 1920 = 96$	**96 KD**
3.9	$\text{before} = 943.5 \div 1.02 = 925$ $943.5 - 925 = 18.5$	**18.5 KD**
3.10	$\text{before} = 1265 \div 1.1 = 1150$ $1265 - 1150 = 115$	**115 KD**
3.11	$\text{before} = 2340 \div 1.04 = 2250$ $2340 - 2250 = 90$	**90 KD**
3.12	$\text{before} = 1428 \div 1.05 = 1360$ $1428 - 1360 = 68$	**68 KD**

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)		
Q	**Working**	**Answer**
finding the formula, then using it backwards		
4.1	**(a)** $m = kr^3$ $24 = k \times 8$ $k = 3$ **(b)** $81 = 3r^3$ $r^3 = 27$ $r = 3$	**(a)** $m = 3r^3$ **(b)** $r = 3$
4.2	**(a)** $m = kr^3$ $54 = k \times 27$ $k = 2$ **(b)** $128 = 2r^3$ $r^3 = 64$ $r = 4$	**(a)** $m = 2r^3$ **(b)** $r = 4$
4.3	**(a)** $m = kr^2$ $80 = k \times 16$ $k = 5$ **(b)** $125 = 5r^2$ $r^2 = 25$ $r = 5$	**(a)** $m = 5r^2$ **(b)** $r = 5$
4.4	**(a)** $m = kr^3$ $16 = k \times 8$ $k = 2$ **(b)** $250 = 2r^3$ $r^3 = 125$ $r = 5$	**(a)** $m = 2r^3$ **(b)** $r = 5$

4.5	**(a)** $m = kr^3$ $320 = k \times 64$ $k = 5$ **(b)** $135 = 5r^3$ $r^3 = 27$ $r = 3$	**(a)** $m = 5r^3$ **(b)** $r = 3$
4.6	**(a)** $m = kr^2$ $36 = k \times 9$ $k = 4$ **(b)** $100 = 4r^2$ $r^2 = 25$ $r = 5$	**(a)** $m = 4r^2$ **(b)** $r = 5$
inverse proportion to x²		
4.7	$y = \frac{k}{x^2}$ $8 = \frac{k}{9}$ $k = 72$ $y = \frac{72}{36} = 2$	$y = 2$
4.8	$y = \frac{k}{x^2}$ $20 = \frac{k}{4}$ $k = 80$ $y = \frac{80}{16} = 5$	$y = 5$
4.9	$y = \frac{k}{x^2}$ $4 = \frac{k}{25}$ $k = 100$ $y = \frac{100}{4} = 25$	$y = 25$

4.10	$y = \frac{k}{x^2}$ $5 = \frac{k}{16}$ $k = 80$ $y = \frac{80}{4} = 20$	$y = 20$
4.11	$y = \frac{k}{x^2}$ $36 = \frac{k}{1}$ $k = 36$ $y = \frac{36}{9} = 4$	$y = 4$
4.12	$y = \frac{k}{x^2}$ $1.5 = \frac{k}{100}$ $k = 150$ $y = \frac{150}{25} = 6$	$y = 6$
direct proportion to √x		
4.13	$y = k\sqrt{x}$ $10 = k \times 2$ $k = 5$ $y = 5 \times 6 = 30$	$y = 30$
4.14	$y = k\sqrt{x}$ $15 = k \times 5$ $k = 3$ $y = 3 \times 8 = 24$	$y = 24$
4.15	$y = k\sqrt{x}$ $6 = k \times 3$ $k = 2$ $y = 2 \times 9 = 18$	$y = 18$
4.16	$y = k\sqrt{x}$ $8 = k \times 4$ $k = 2$ $y = 2 \times 10 = 20$	$y = 20$

4.17	$y = k\sqrt{x}$ $14 = k \times 2$ $k = 7$ $y = 7 \times 3 = 21$	$y = 21$
4.18	$y = k\sqrt{x}$ $30 = k \times 6$ $k = 5$ $y = 5 \times 2 = 10$	$y = 10$
what happens to y?		
4.19	$3^2 = 9$	y **is multiplied by 9**
4.20	$2^2 = 4$	y **is divided by 4**
4.21	$2^3 = 8$	y **is multiplied by 8**
4.22	x is multiplied by 5	y **is divided by 5**
4.23	$\sqrt{4} = 2$	y **is multiplied by 2**
4.24	$10^2 = 100$	y **is divided by 100**